HAREESH REDDY EMANI

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Objective - Internship/Co-op in Digital & RTL Design, Verification starting Summer 2025.

Education

Portland State University, Oregon, USA **Sep 2024 - Jun 2026**

MS in Electrical and Computer Engineering **CGPA – 3.6/4.0**

Coursework–Introduction to System Verilog, Formal Verification, ASIC Design and Synthesis, Microprocessor System Design, Verilog Workshop, Computational Tools

Vasireddy Venkatadri Institute of Technology **Aug 2020 - Jul 2024**

Bachelor of Engineering - Electrical, Electronics and Communications Engineering **CGPA – 7.92/10.0**

Technical Skills

Programming: System Verilog, Verilog, C
Tools: QuestaSim, ModelSim, VC Formal, Matlab
Scripting Language: Python
Operating Systems: Windows, Linux
Protocols: APB, SPI, UART, MESI
Concepts: Digital logic Design, RTL, DDR4**,** Caches, assertions, Object Oriented Programming, FSM, Randomization & Constraints

Academic Projects

AMBA APB Interface Design and Verification [System Verilog]

- Designed an AMBA APB (Advanced Peripheral Bus) interface using System Verilog.
- Uses a finite state machine (FSM) for protocol control.
- Implemented address, write, and read operations as per the APB protocol.
- Developed a System Verilog testbench to verify functional correctness.
- Simulated and tested the design to ensure protocol compliance.

Design and Formal Verification of a Sequence Detector FSM for Enhanced Security Applications

- Designed sequence detector FSM using System Verilog for security enhancements.
- Applied Synopsys VC Formal for rigorous FSM verification and reliability.

SPI (Serial Peripheral Interface) Controller Design and Verification [System Verilog]

- Implements SPI Master with configurable clock polarity and phase.
- Developed RTL design for SPI master and slave modules.
- Verified using a SystemVerilog testbench.

Design of Cache Controller using MESI protocol [System Verilog]

- Simulated the behavior of cache controller for 32-bit processor backed by L2 by maintaining inclusivity.
- Write back and write allocate policies used, True LRU eviction to replace blocks and MESI to maintain coherence.

Design and Verification of Asynchronous FIFO **[System Verilog]**

- RTL design for FIFO aiming for asynchronous read and write. Prepared Test Plan and Architected Test bench.
- Implemented synchronizers and used gray code pointers to reduce the probability of metastable states.

Training

February 2024 – August 2024

Lucid VLSI, Hyderabad, India - Advanced VLSI Design and Verification.
Concepts: Advanced Digital Design using Verilog, System Verilog
Tools Used: ModelSim